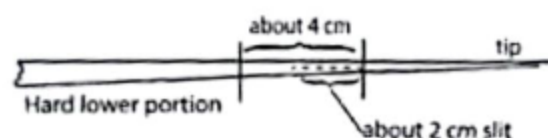


1 You are provided with the following materials:

- Two similar leaves of *Brassica oleraceae* (Sukuma wiki)
- A scalpel
- 5 cm³ of liquid K₁ in a test tube
- 5 cm³ of liquid K₂ in a test tube
- (Access) to means of timing

Procedure

- (i) Remove the entire leafy parts along the midribs of both leaves.
- (ii) Retain the two midribs still attached to their petioles.
- (iii) Discard the hard lower petiole.
- (iv) Measure about 4 cm of the remaining midrib towards the tip. Cut and discard the tip. The process is illustrated as follows:



- (v) Make a 2 cm slit from the tip end of each of the 4 cm portions as shown in the diagram above.
- (vi) Place one piece into the test tube with liquid K₁ and the other into liquid K₂ and leave them for 20 minutes. Remove the two pieces and make observations.
- (a) Draw the appearance of each piece.

- (i) Piece from K₁



(1 mark)

- (ii) Piece from K₂

(1 mark)

(b) Account for the observations made on the piece from each liquid.

(i) Piece from K_1

(3 marks)



(ii) Piece from K_2

(3 marks)

(c) State how the experiment would be modified to obtain the same results within a shorter period of time. (2 marks)

(d) Explain why the petiole and the lower parts of the midribs were not suitable for use in this experiment. (1 mark)

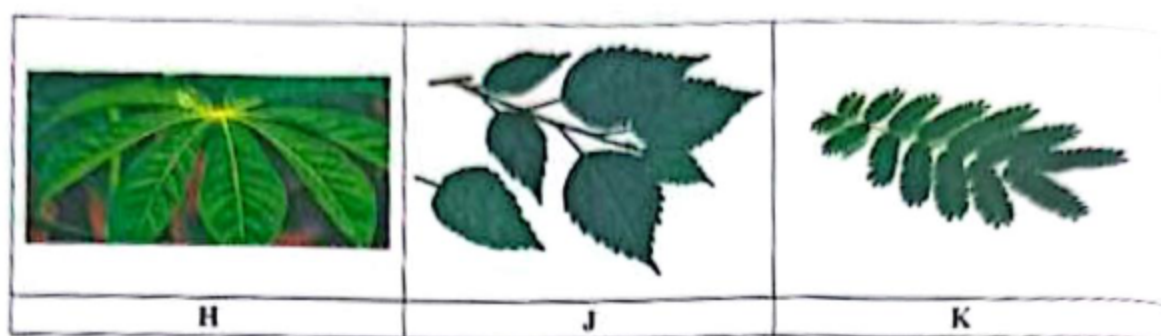
2

You are provided with three plant specimens labelled E, F and G obtained from different plants belonging to different Families.

(a) Use the specimens provided together with the photographs below to construct a dichotomous key that can be used to identify them. Use the features below in the order given to construct the key: (10 marks)



- Simple or compound leaves
- Leaf venation
- Type of compound leaf
- Leaf margin
- Nature of leaf lamina



- (b) Fill the following table indicating the steps followed to identify specimens E, F and G. (3 marks)

Specimen	Steps
E	
F	
G	

- (c) (i) State **one** feature in the root and **one** in the stem of specimen G that places the plant in its Class. (2 marks)

Root

Stem

- 3 You are provided with the following materials:

- 3 test tubes and means of labelling them
- Solutions L_1 , L_2 and L_3 ,
- 10 cm³ measuring cylinder,
- Iodine solution.

Procedure

- Label the three test tubes A, B and C.

3 You are provided with the following materials:

- 3 test tubes and means of labelling them
- Solutions L_1 , L_2 and L_3 ,
- 10 cm³ measuring cylinder,
- Iodine solution.

Procedure

- Label the three test tubes A, B and C.
- To test tube A, add 1cm³ of L_1 , add one drop iodine solution. Record the observations in the table below.
- Add 1cm³ of each of L_1 and L_2 into tube B. Place it on the test tube rack and leave it undisturbed for ten minutes. Add a drop of iodine solution and record the observations in the table below.
- To the third test tube, C, add 1cm³ of L_2 , add two drops of dilute hydrochloric acid. Leave the contents undisturbed for ten minutes. Add 1cm³ of L_1 , shake the contents and again place the contents on the test tube rack for about five minutes, add a drop of iodine solution.
- Record the observations and inferences in the table below.

Test tube	Observations after adding iodine solution	Conclusion
A		
B		
C		

(6 marks)

- (a) (i) Suggest the likely identity of solution L_2 .

(1 mark)

- (ii) Explain your answer in 3(a)(i) .

(2 marks)



- (b) Suggest with a reason where the process being investigated in this experiment would take place in the human alimentary canal.

(1 mark)

(i) Part of alimentary canal

(ii) Reason (2 marks)

- (c) State two other modifications one would make in test tube C to obtain similar observations

(2 marks)

